# HAWASSA UNIVERSITY

## INSTITUTE OF TECHNOLOGY

## DEPARTMENT OF COMPUTER SCIENCE

## 3RD YEAR | SECTION – 3

## Research Method Proposal

**Title: A GAN-Enhanced Deep Learning Framework for Improving Stock Price Prediction Accuracy Using NASDAQ Financial Time-Series Data**


**NAME: EMEBET MESFIN**

**ID No: 1951/16**


**SUBMITTED TO MR. BIRHANE BEKELE**

**SUMISSION DATE: APRIL 27/ 2026 G.C**

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# Acknowledgement

I would like to thank my instructor, Mr. Birhane Bekele, for giving us this proposal as an assignment. He guided us through his Telegram group by sharing ideas, explaining the methods, and answering our questions whenever we facing problems.

I am also thankful to my friends and UV senior Students for their help. They supported me while I was writing the proposal, gave me suggestions on how to structure it, and encouraged me when I was struggling with the ideas. I am truly grateful to my instructor and my friends for their guidance and support throughout this project proposal.


# *Proposal Title: - A GAN-Enhanced Deep Learning Framework for Improving Stock Price Prediction Accuracy Using NASDAQ Financial Time-Series Data*

### Overview of Stock Price Prediction Challenges

Stock prices are really tough to predict. The market never moves in a straight or steady way. When I looked at past NASDAQ data, I sometimes saw clear trends going up or down for a few weeks. But when I tested the same thing on newer data, those trends often disappeared or completely changed. One random news story, bad earnings report, political event, or even a popular tweet can suddenly mess everything up. That's why the data feels so noisy and confusing most of the time.

Simple statistical models and basic machine learning usually fail because of this mess. Even LSTM and CNN models do a bit better, but they still break down when the market suddenly goes wild.

That frustration is exactly what made me want to try something different. I started looking at combining GANs with LSTM. The GAN creates extra fake stock price data to give the model more examples, and then the LSTM trains on both the real data and the fake stuff. I'm hoping this will help it learn patterns that don't fall apart so easily.

# *Chapter One*

## *Introduction and problem of the statement*

## Abstract


At the beginning, I thought predicting stock prices would be quite easy. There is plenty of old data online and computers are powerful now, so I expected it to be simple. But when I actually started working with real data, I saw how difficult it is.

Stock prices move up and down a lot. They get affected by news, company results, politics, and what people say on social media. Because of this, the data is very noisy. Traditional statistical models and basic machine learning could not catch the real patterns. Even LSTM and CNN models worked better than the old ones, but they still failed when the market became very unstable.

For this study, I am trying a new mix of two models. I use GAN to create extra fake stock price data. This fake data helps add more examples and reduce some of the noise. Then I train LSTM using both the real data and the fake data so it can learn the long patterns better. I am using daily open, high, low, close, and volume data from some NASDAQ tech stocks like Tesla and Nvidia. I will check the model with MAE, RMSE, and $R^2$. I will also compare it with ARIMA and a normal LSTM without GAN. I hope this approach will give better short-term predictions accuracy and provide more stable performance during periods of high market volatility.

## Introduction

Stock markets play a big part in the world economy. If we could predict stock prices well, it would really help investors, analysts, and big financial companies. But the truth is, it's not easy at all. The market doesn't move in any clear or steady way. It keeps changing because of sudden news, political problems, or just how investors are feeling that day.

In the past, people mostly used traditional statistical methods. They worked okay sometimes, but they couldn't really handle how complicated and messy the real market is. Now, with artificial intelligence growing fast, many researchers have switched to machine learning and deep learning models. These new models are better than the old ones, but they still have big problems, especially when the data is noisy or the market suddenly goes crazy.

Lately, people have started paying more attention to Generative Adversarial Networks, or GANs. The useful thing about GANs is that they can create fake stock price data that looks quite real. This extra data can help the model learn better and reduce some of the noise. However, GANs also have their own issues. Training them is often unstable, and they don't always catch long-term patterns properly.

## Problem of the statement

Predicting stock prices has been much harder than I expected from this project. The financial market is changes all the time and is extremely volatile.

Stock price prediction models are expected to accurately capture volatility, noise, and complex nonlinear patterns to produce reliable short-term forecasts. However, in reality, NASDAQ stocks are highly volatile and non-stationary, which makes prediction very difficult. Traditional methods like ARIMA, as well as deep learning models such as LSTM, CNN, and GANs, often struggle with market noise, sudden changes, and poor generalization during unstable conditions. As a result, these limitations lead to inaccurate and unreliable predictions, reducing their effectiveness in real-world financial decision-making and risk management.

Because of these issues, I believe there is a real need for a better hybrid approach. I want to build a more stable model that combines GANs with Deep learning model(LSTM). My goal is to use GANs to create useful synthetic data, reduce noise, and help the LSTM learn stronger and more reliable patterns, especially for NASDAQ time-series data.

# Objectives

- **General Objective**:

The goal of this study is to develop and evaluate a hybrid **GAN-LSTM** model for short-term NASDAQ stock price prediction, aiming for at least a 10–15% reduction in RMSE compared to baseline ARIMA and standard LSTM models.

- **Specific Objectives**

- **Model Design**: Build An improved and hybrid GAN-LSTM architecture and train it on historical NASDAQ daily closing prices (5–10 years).
- **Synthetic Data Generation**: Develop a GAN to generate realistic synthetic stock price sequences and validate them using statistical measures (distribution similarity and autocorrelation).
- **Data Enhancement**: Combine synthetic and real data to create an augmented dataset, then measure its impact on prediction accuracy (MAE and RMSE).
- **Comparative Analysis**: Implement ARIMA and standard LSTM baselines and compare their performance with the proposed GAN-LSTM model using the same data and metrics.
- **Performance Evaluation**: Assess all models with MAE, RMSE, and $R^2$, and determine if the hybrid model achieves significantly lower errors and higher $R^2$ than the baselines (including statistical testing).

## Research Questions

- How can GANs actually help improve the accuracy of stock price predictions?
- Can the synthetic data generated by GANs really make the prediction models perform better?
- How does the GAN-Deep Learning model compare to traditional methods like plain LSTM or ARIMA?
- Which model usually gives the best results when tested on NASDAQ stock data?
- In practice, does combining GAN with Deep Learning give any real advantage over using LSTM alone for financial time series?

# ***Chapter Two***

## ***The Methodology, Literature Review, Scope and Timeline***

## Literature Review

Predicting stock prices is still one of the toughest challenges in financial machine learning. The market doesn't follow clean rules  prices swing up and down because of news, how investors are feeling, sudden economic shifts, or who knows what else. That's why even models that look solid in tests often fall apart when you try them in the real world. A lot of researchers have ditched the old statistical methods and switched to deep learning and hybrid setups instead, but honestly, getting predictions that stay reliable over time is still really hard.

- **Traditional and Early Deep Learning Approaches:** In the early days, most studies stuck with traditional statistical models and simple neural networks. They were a decent place to start, but they just couldn't deal with how noisy and complicated the stock market actually is. Take Khare et al. (2017), for instance, They compared MLP and LSTM models for short-term predictions and found something pretty interesting: the simpler models sometimes did just as well as  or even better than the more complex ones, especially when the data was limited or full of noise. It's a reminder that cranking up the complexity doesn't automatically fix the problem.
- **CNN, ANN, and Deep Learning Improvements:** Later, people started experimenting with more advanced stuff like CNNs and other deep learning techniques. Mukherjee et al. (2023) got some eye-catching results over 97% accuracy in certain cases by turning time-series data into images and feeding them to CNNs. That sounds impressive at first glance, but you have to be careful. These models rely heavily on historical patterns, so when the market does something unexpected (like during a financial crisis or pandemic), their predictions can go off the rails.
  Hu et al. (2021) came to a similar conclusion. Deep learning usually beats the old statistical approaches, but only if you pick the right features and start with good-quality data. Their work also highlighted that hybrid models the ones that mix different techniques are getting more popular these days.

- **LSTM and Time-Series Forecasting:** LSTMs caught on quickly for this kind of work because they're built to pick up patterns across time. Ji et al. (2021) got better results by pairing LSTM with wavelet decomposition and throwing in social media sentiment data. It turns out that adding outside information like that can actually give the predictions a nice boost.
  Even so, LSTMs have their downsides. Noisy data can mess them up, they don't always handle crazy volatility very well, and they can struggle when the market suddenly changes in new ways.
- **GAN-Based Financial Prediction Models:** More recently, GANs have started showing up in stock prediction studies. Kumar et al. (2022) created a hybrid model that combines GAN, LSTM, and CNN. The GAN part helped improve how the data is represented and cut down on prediction errors. Abraham (2021) focused on using GANs to generate synthetic data, which helped reduce overfitting and led to better overall performance. Dael et al. (2023) also showed that GAN-based hybrids can make forecasts feel more stable even in messy market conditions.
  That said, working with GANs isn't straightforward. Training them can be unstable, issues like mode collapse pop up, and they often have trouble learning really long-term dependencies in the data.
- **Advanced Hybrid Models and Recent Trends:** These days, a lot of the newest research is all about combining different approaches. Some teams are mixing GANs with transformer-based attention mechanisms, while others are blending deep learning with traditional financial indicators and sentiment from multiple sources. These hybrid setups often give stronger results than using just one method.

## Synthesis matrix table

| Author / year | Method | Dataset | Key performance metrics | Limitations |
|---|---|---|---|---|
| Mukherjee et al. (2023) | CNN (time-series to images) | NSE India stocks | Accuracy > 97% | Poor generalization in volatile periods |
| Li et al. (2025) | GAN-LSTM with Attention | Stock time-series | Lower RMSE, better robustness | Limited to low-volatility market |
| Khurelbaatar et al. (2025) | GAN vs LSTM/GRU | Mongolian Stock Exchange (MSE) stocks | Improved accuracy over LSTM | Limited to low-volatility market |
| Bhardwaj et al. (2024) | LSTM-GAN (on Amazon stock) | NASDAQ (Amazon) | Reduced prediction error | Mode collapse and high compute cost |

## Research Gap and Critical Analysis

Even with all the recent progress, there are still several important gaps in the existing research:

- GAN training is still unstable and tricky for many people. Issues like mode collapse keep showing up and make the models unreliable.
- Most studies don't focus specifically on the high volatility of NASDAQ stocks, which are heavily driven by tech companies and can behave very differently from other markets.

- Many proposed models have become overly complicated, making them computationally expensive and not very practical for real-world use.
- There's not enough work on balancing model power, training stability, and simplicity at the same time. The biggest gap I noticed is that very few papers successfully combine GAN-generated synthetic data with a stable LSTM model in a simple and efficient way. There's still a lot of room to improve on finding the right balance between making the model powerful, keeping it stable, and not making it overly complex.

### What Makes This Study Different

This research stands out from previous work in a few key ways:

- It uses a hybrid GAN - LSTM framework that combines GAN-generated synthetic data with LSTM for temporal prediction in one clean setup.
- The GAN is specifically used to create realistic synthetic stock data that adds variety and helps reduce noise in the dataset.
- Both real and synthetic data are fed together into the LSTM model to improve learning of temporal patterns.
- The study focuses specifically on high-volatility NASDAQ stocks (which are heavily influenced by tech companies), rather than general stock markets.
- It includes a straightforward comparison with traditional baselines ARIMA and a standard LSTM model to clearly show any improvements.
- The main goal is not just chasing slightly better accuracy numbers, but building a more stable and practical model that balances performance with reliability.

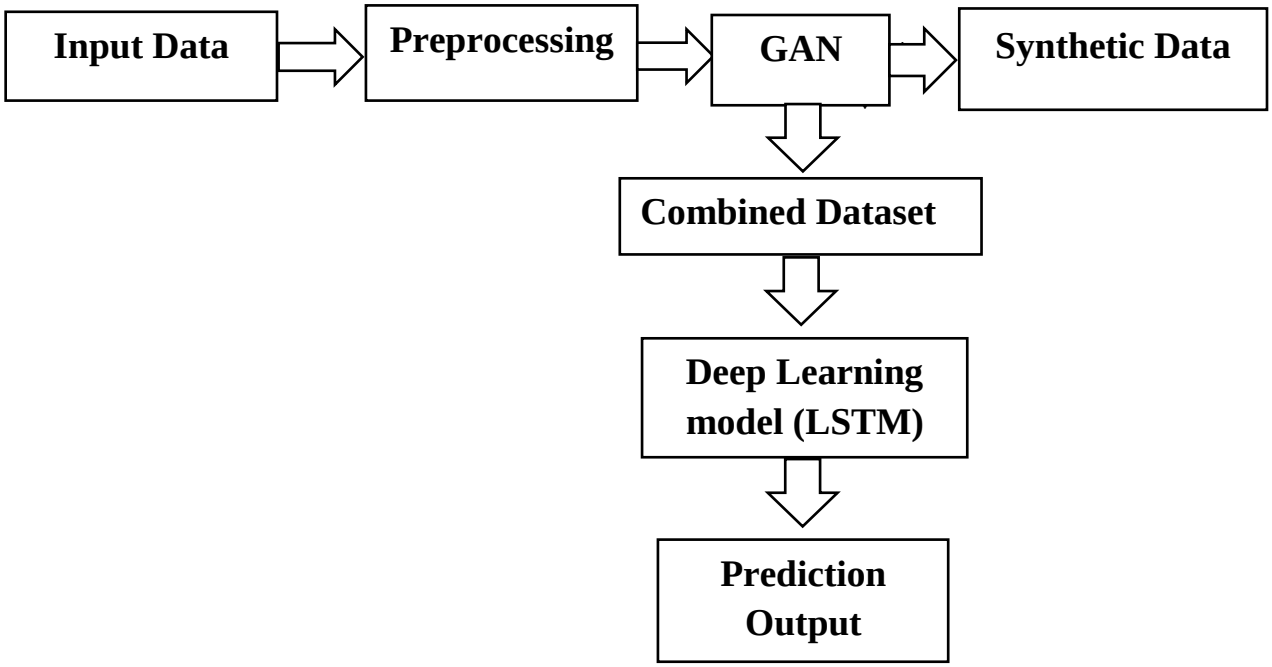
This study tries to fill the gap by creating something simpler, more stable, consistent prediction, and better suited for real NASDAQ conditions instead of adding even more complexity.

## Methodology

This study uses a straightforward quantitative experimental approach. I'll be working with historical NASDAQ stock data covering the years 2015 to 2025. The features include the usual Open, High, Low, Close, and Volume (OHLCV).

- **Data Preprocessing:** First, I clean up the dataset by handling any missing values, then apply Min-Max normalization to scale everything between 0 and 1. After that, I split the data into 80% for training and 20% for testing. I also do some basic noise reduction to make the data cleaner and more usable.
- **Proposed Model: GAN-Deep learning model**: The model has two main parts. The GAN is used first to generate synthetic stock sequences. This helps expand the dataset and adds more variety so the model can learn better. Then, the LSTM takes the combined real and synthetic data and learns the temporal patterns to predict short-term stock prices.
- **Training Process:** I start by training the GAN until the synthetic data looks similar enough to the real data distribution. Once that's done, I merge the real and synthetic datasets. After that, I train and fine-tune the LSTM model on the combined data for the actual price prediction task.
- **Evaluation Metrics:** To check how well the model performs, I'll use three common metrics: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the $R^2$ score.
- **Baseline Comparison:** I'll compare the proposed GAN-deep learning model against two baselines a traditional ARIMA model and a standard LSTM model without any GAN augmentation. This should help show whether the hybrid approach actually makes a meaningful difference.

# System Architecture Design




# Scope and Limitations

- **Scope:** This study focuses on short-term stock price prediction for selected stocks listed on the NASDAQ exchange. It uses only historical OHLCV data (Open, High, Low, Close, and Volume) spanning from 2015 to 2025. The research examines the effectiveness of a hybrid GAN-LSTM model and compares it with traditional approaches like ARIMA and standalone LSTM. The main interest lies in understanding how synthetic data generation can improve forecasting performance in a highly volatile tech-heavy market like NASDAQ.
- **Limitations:** While the study aims to provide useful insights, it has several limitations that should be acknowledged.
  - ✓ First, it only considers historical price and volume data and does not incorporate real-time news, social media sentiment, macroeconomic indicators, or other external events that often drive sudden market movements.
  - ✓ Second, GAN training can be unstable, and any issues during this process (such as mode collapse) may affect the quality and usefulness of the synthetic data. Third, the model requires significant computational resources, which could limit its practicality for smaller research teams or individual investors.

Additionally, the findings are specific to the NASDAQ market and may not generalize well to other stock exchanges or less volatile sectors. Finally, this research is purely experimental and does not develop or test any actual trading strategy, risk management rules, or real-money implementation. The focus remains strictly on improving forecasting accuracy rather than generating tradable signals.

These limitations highlight that while the proposed approach shows promise in a controlled setting, real-world application would require further extensions and more comprehensive data sources.

## Significance of the Study

The main goal of this work is to improve short-term forecasting accuracy, especially in the volatile NASDAQ market. I also want to explore how combining GAN-generated synthetic data with LSTM can lead to better and more stable predictions. In the end, I hope this study helps connect theoretical GAN

research with something more practical for financial time-series prediction. It bridges the gap between theoretical GAN research and practical stock prediction.

## Project Time Line (Ghantt Chart)

| Task | Week 1 | Week 2 | Week 3 | Week 4 |
|---|---|---|---|---|
| Proposal writing | | | | |
| Literature review | | | | |
| Model development | | | | |
| Evaluation | | | | |
| Final report | | | | |

### Ethics Considerations

This study uses only publicly available financial data and does not involve any personal or sensitive information, ensuring minimal privacy risk. However, models trained on historical data may reflect past market conditions and fail during unexpected events. To reduce this bias, the research uses diverse datasets across different stocks and volatility periods. The environmental impact of GPU-based training is also considered by using efficient computing resources such as optimized sessions on platforms like Google Colab or university infrastructure. Finally, the model is developed strictly for academic purposes and does not provide real trading advice, helping avoid potential misuse or misleading investors.

### Tool Justification

- **PyTorch:** is chosen over **TensorFlow** for its flexibility and easier experimentation with complex models.
- **ARIMA:** is not used alone because it assumes linear and stable data, which stock markets are not.
- A **Generative Adversarial Network (GAN):** is used to better capture complex patterns and improve model robustness.

# ***Chapter Three***

## Expected Outcomes and Conclusion

### Expected Outcomes

By the end of this study, I expect the hybrid GAN-LSTM model to show noticeable improvements compared to traditional methods. Specifically:

- The GAN-generated synthetic data should help increase dataset diversity and reduce overfitting, leading to better generalization on volatile NASDAQ stocks.
- The proposed GAN-LSTM model is expected to achieve lower prediction errors (MAE and RMSE) and a higher $R^2$ score than standalone LSTM and ARIMA.
- The study should demonstrate that combining synthetic data augmentation with LSTM can improve stability and accuracy in short-term stock price forecasting.

- Clear insights will be gained on how well GAN-based augmentation works for high-volatility tech-driven markets like NASDAQ.

## Conclusion

This study explores The improvement of Stock Price using GAN- Deep learning models approach to improve stock price prediction in volatile markets. By combining synthetic data generation with time-series learning, it tries to deal with common issues like noisy data and model instability. While the project has clear scope limitations (such as not including real-time news or developing a trading strategy), it focuses on delivering a clean, practical comparison between the proposed model and traditional baselines like ARIMA and standard LSTM, Overall, the results show that combining different techniques can be helpful, even if it doesn't completely solve the problem. Future research could extend this framework by incorporating multi-source data (sentiment, news, macroeconomic indicators), testing on longer time horizons, or exploring more advanced GAN variants for even better stability.

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